



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Microprocessors and Microcontrollers
2	Course Code	CSE 4325
3	Trimester and Year	Fall 2025
4	Pre-requisites	CSE 3313 Computer Architecture
5	Credit Hours	3
6	Section	
7	Class Hours	
8	Class Room	
9	Instructor's Name	Arnab Bhattacharjee (ArBh)
10	Email	arnab@cse.uiu.ac.bd
11	Office	Room 933(I)
12	Counselling Hours	Updated in eLMS
13	Text Book	1. The Intel Microprocessors by Barry B.Brey [8th edition] 2. Microprocessor and Microprocessor System Design by Dr. M Rafiquzzaman [2nd edition]
14	Reference	1. Assembly language programming by Ytha Yu, Charles Marut 2. Microprocessor and Interfacing by Douglas V Hall 3. Digital computer electronics by Albert Paul Malvino
15	Course Contents (approved by UGC)	Introduction to microcomputer architecture, 16-bit, and 32-bit microprocessors: architecture, addressing modes, instruction set, interrupts, multitasking and virtual memory, paging; multicore processor, cache memory, interfacing: programmable peripheral interface, direct memory access, keyboard and display interface, memory chips (i.e. ROM, RAM), Clock generator, Bus Arbiter; Microcontroller: Introduction to micro- controllers (i.e. ATmega32), Digital and analog signals and conversions, Interfacing switches, LEDs, sensors and motors, serial communication protocols (i.e. SPI, I2C, CANBUS) in embedded system.

16	Course Outcomes (COs)						
	COs	Description	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Engineering Problem Solving	Complex Engineering Activities
	CO1	Identify the components of microprocessor architecture and its functionalities	C	a <i>Engineering Knowledge</i>	K3 <i>Engineering fundamentals</i>	P1 <i>Depth of Knowledge</i>	-
	CO2	Implement microprocessor interfacing techniques with memory devices and external peripherals.	C	b <i>Problem Analysis</i>	K4 <i>Specialist knowledge</i>		
	CO3	Discuss operating mode of microprocessors, R/W timing diagram, microprocessors memory management techniques.	C	a <i>Engineering Knowledge</i>			
	CO4	Explain SAP (simple as possible) computer architecture.	C	a <i>Engineering Knowledge</i>			
	CO5	Identify the difference between micro-processor and micro-controller, use analog to digital conversion (ADC) and digital to analog conversion (DAC) using micro-controller and apply different communication protocol for interfacing.	C	b <i>Problem Analysis</i>			

17	Teaching Methods	Lecture, Lecture Slides, Case Studies, Project Developments.
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18	CO with Assessment Methods	<table><tr><th>CO</th><th>Assessment Method</th><th>(%)</th></tr><tr><td>-</td><td>Attendance</td><td>5</td></tr><tr><td>-</td><td>Assignments</td><td>5</td></tr><tr><td>-</td><td>Class Tests</td><td>20</td></tr><tr><td>CO1, CO2</td><td>Midterm exam</td><td>30</td></tr><tr><td>CO3, CO4, CO5</td><td>Final exam</td><td>40</td></tr></table>	CO	Assessment Method	(%)	-	Attendance	5	-	Assignments	5	-	Class Tests	20	CO1, CO2	Midterm exam	30	CO3, CO4, CO5	Final exam	40																																																																																		
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			Mohamad Rafiquzzaman >Chapter 3 section 3.1, 3.2, 3.3 Assembly language programming Ytha Yu > Chapter 3 section 3.2	functionality of the pins.
4	Memory segmentation	CO1	Microprocessors and microcomputer-based system design Mohamad Rafiquzzaman >Chapter 3 section 3.1, 3.2, 3.3	Describe the memory organization of microcomputer system.
5	Intel 8086 interfacing with ROM and RAM	CO2	Microprocessors and microcomputer-based system design Mohamad Rafiquzzaman >Chapter 1 section 1.3	Describe the Intel 8086 interfacing with ROM and RAM.
6	Programmable Peripheral Interface 82C55	CO2	The Intel Microprocessors 8th Edition by Barry B Brey >> 11–3 The Programmable Peripheral Interface page 395 Microprocessor and Interfacing, Programming of Hardware” by Douglas Hall chapter 9 page 261	Describe the programmable peripheral interface (82C55) modes.
7	Intel 8086 interrupt mechanism	CO1	The Intel Microprocessors 8th Edition	Describe the Intel 8086

			by Brey >6-4 Introduction to Interrupts: 8086 Interrupt	interrupt mechanism.
		MIDTERM EXAM		
8	Intel 8086 mode of operation	CO3	The Intel Microprocessors 8th Edition by Brey > Section 9-6 Minimum Mode versus Maximum Mode page 323 & Bus Controller Section 13-3 Shared-bus operation 506	Explain the Intel 8086 mode of operation.
9	Intel 8086 bus timing diagram	CO3	The Intel Microprocessors 8th Edition by Brey > Section 9-4 Bus Timing page 315	Explain the Intel 8086 bus timing diagram for read and write operation.
10	Intel 8086 pipelining mechanism	CO3	8086 Pipelining.pptx	Describe the Intel 8086 pipelining mechanism.
11	Operation of clock generator 8284A	CO3	The Intel Microprocessors 8th Edition by Barry B Brey > Section 9-2 Clock generator page 307	Describe the operation of clock generator 8284A and interface with Intel 8086
12	Introduction to protected mode addressing and paging	CO3	The Intel Microprocessors 8th Edition by Barry B Brey >>2-2 Real mode memory addressing page 58 2-3 Introduction to protected mode memory addressing page 63 2-4 Memory paging page 68	Explain the difference between real mode addressing and protected mode addressing.
13	SAP-1 (simple as possible) computer architecture	CO4	Digital computer electronics by Albert Paul	Explain SAP1 architecture and

			Malvino >> Part 2 SAP page 140	programming
14	Introduction to Micro-controller	CO5	microcontroller note1.pptx	Explain the difference between microprocessor and micro- controller
15	ATMega32 architecture	CO5	microcontroller note2.pptx	Describe the memory organization of micro-controller system.
16	Analog to digital conversion (ADC) and digital to analog conversion (DAC)	CO5	analog-digital.pptx	Explain analog to digital conversion (ADC) and digital to analog conversion (DAC) using micro- controller.
17	Micro-controller serial protocols	CO5	Communication Protocols.pptx	Explain the microcontroller communication protocols SPI, I2C, CANBUS protocol.

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5%
Assignments	5%
Class Tests	20%
Mid Term	30%
Final Exam	40%

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve complex engineering problems
PO3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	An ability to investigate complex problems using research-based knowledge and research methods design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
PO8	An understanding of professional and ethical responsibility
PO9	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	An ability to communicate effectively
PO11	Project management and finance
PO12	A recognition of the need for, and an ability to engage in life-long learning